

ORANGE COUNTY AP, VA, USA

WMO: 722167

Lat: 38.247N

Lon: 78.046W

Elev: 143

StdP: 99.62

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03718

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.6	-17.9	0.8	-6.2	-15.3	1.0	-3.7	9.2	8.6	8.3	6.4	0.9	310	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	34.0	24.4	32.6	24.2	31.3	23.8	27.0	30.2	25.8	30.1	25.1	29.6	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.9	28.3	24.9	20.4	27.7	23.7	18.8	26.8	85.3	30.2	80.4	31.0	77.1	28.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.7	5.5	DB	-14.8	36.5	3.0	1.5	-17.0	37.6	-18.8	38.5	-20.4	39.3	-22.6	40.4	(4)
(5)			WB	-15.3	27.5	2.9	1.1	-17.4	28.3	-19.1	28.9	-20.8	29.5	-22.9	30.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	2.1	3.4	7.5	13.3	18.3	22.7	25.0	24.1	20.7	14.3	8.1	4.1	(6)
(7)		DBStd	9.22	5.45	5.44	5.58	4.81	4.27	3.05	2.52	2.59	3.49	4.59	4.60	4.85	(7)
(8)		HDD10.0	880	250	194	119	21	1	0	0	0	13	90	193		(8)
(9)		HDD18.3	2388	502	418	337	164	57	3	0	16	141	309	441		(9)
(10)		CDD10.0	2228	6	10	42	119	257	382	466	437	320	147	32	11	(10)
(11)		CDD18.3	692	0	0	1	12	54	135	208	179	86	17	1	0	(11)
(12)		CDH23.3	6402	0	1	27	187	533	1238	1966	1582	719	142	8	0	(12)
(13)		CDH26.7	2456	0	0	1	48	155	457	863	640	253	38	1	0	(13)
(14)	Wind	WSAvg	1.7	2.1	2.2	2.3	2.3	1.7	1.5	1.3	1.1	1.3	1.6	1.7	1.7	(14)
(15)	Precipitation	PrecAvg	1091	74	74	89	78	99	107	116	96	106	88	93	77	(15)
(16)		PrecMax	1602	234	191	213	238	222	309	290	300	324	318	307	180	(16)
(17)		PrecMin	776	7	9	8	18	21	16	25	12	9	0	12	8	(17)
(18)		PrecStd	180	43	46	44	44	47	69	62	56	78	67	59	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.2	26.3	30.9	32.1	34.9	36.9	35.1	34.3	31.0	24.2	21.2	(19)
(20)			MCWB	16.9	15.5	16.2	20.3	23.8	25.4	25.8	24.9	24.1	22.2	17.7	17.5	(20)
(21)		2%	DB	17.1	18.0	23.0	27.8	30.2	32.6	34.8	32.9	32.2	27.3	21.2	17.7	(21)
(22)			MCWB	13.0	12.4	14.6	18.0	22.2	24.3	24.9	24.4	23.3	20.2	15.6	15.0	(22)
(23)		5%	DB	14.0	15.3	20.9	25.9	28.0	31.2	32.7	32.1	29.9	24.7	18.6	15.0	(23)
(24)			MCWB	10.6	9.6	13.7	16.9	21.2	23.7	24.5	24.3	22.3	18.6	14.0	11.9	(24)
(25)		10%	DB	11.0	12.4	17.5	22.5	27.0	29.8	31.3	30.3	27.4	22.3	16.8	12.2	(25)
(26)			MCWB	8.0	8.4	12.0	15.7	20.5	22.7	24.2	23.6	21.2	17.9	12.7	9.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	16.4	18.2	21.6	25.1	27.7	28.2	28.1	27.7	23.4	19.5	19.1	(27)
(28)			MCD B	18.6	19.7	23.0	25.8	30.2	29.9	31.0	30.5	30.0	28.0	22.8	19.9	(28)
(29)		2%	WB	14.2	13.8	16.6	19.9	23.5	25.6	26.9	26.7	24.9	22.1	17.2	16.0	(29)
(30)			MCD B	16.6	17.3	21.2	25.6	28.2	30.6	31.0	29.7	28.3	25.7	19.0	17.4	(30)
(31)		5%	WB	11.0	11.3	14.9	18.3	22.5	24.7	25.8	25.6	23.5	20.9	15.4	12.6	(31)
(32)			MCD B	13.2	14.2	19.0	23.6	26.8	29.7	30.6	29.5	27.5	22.8	18.2	14.1	(32)
(33)		10%	WB	8.0	8.5	12.8	16.8	21.4	23.7	25.1	24.7	22.5	19.0	13.0	9.9	(33)
(34)			MCD B	10.4	12.0	17.2	21.1	25.0	28.1	29.8	28.8	26.0	21.9	15.1	11.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	10.9	12.1	13.2	12.0	11.5	11.3	11.1	11.2	11.9	11.8	10.1	(35)
(36)			MCD BR	14.6	15.5	16.5	16.8	14.2	13.4	13.3	13.1	14.4	14.4	15.5	13.2	(36)
(37)			MCWBR	11.3	10.5	9.7	8.2	7.1	5.9	5.0	5.3	6.1	8.0	9.9	10.1	(37)
(38)		5% WB	MCD BR	12.7	13.5	14.5	14.2	12.4	12.0	11.5	10.9	11.1	10.9	12.0	11.3	(38)
(39)			MCWBR	10.9	10.4	9.7	7.9	6.7	6.1	5.6	5.6	5.9	6.6	9.1	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.314	0.349	0.392	0.435	0.482	0.503	0.507	0.412	0.353	0.319	0.302	(40)	
(41)		taud	2.512	2.449	2.398	2.304	2.205	2.114	2.081	2.047	2.299	2.461	2.512	2.531	(41)	
(42)		Ebn at Noon	891	917	911	887	850	807	786	772	834	860	855	859	(42)	
(43)		Edh at Noon	77	94	109	127	143	157	161	163	120	92	77	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.20	3.00	3.99	4.98	5.57	6.12	5.89	5.30	4.37	3.37	2.48	1.94	(44)	
(45)		RadStd	0.18	0.34	0.26	0.38	0.51	0.42	0.32	0.26	0.51	0.38	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (64 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+78		

Nomenclature: See separate page